

SHUBHAM LILHARE

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HARDWARE DESIGN ENGINEER | POWER ELECTRONICS | EMBEDDED DESIGN | MEDICAL & AUTOMOTIVE

Hardware Design Engineer with 5.5+ years of experience in Analog and Power Electronics design across Medical and Automotive domains. Proven expertise in 48V Battery Management Systems, Li-Ion charging systems, DC-DC converters, and board-level hardware architecture. Strong hands-on experience in schematic design (OrCAD/Altium), simulation (LTspice/PSpice), PCB design review, board bring-up, EMI/EMC testing, and design verification. Experienced in leading cross-functional teams and delivering products from concept to validation.

KEY TECHNICAL HIGHLIGHTS

- Designed and validated **4kW (48V, 80A) Battery Management System** for EV applications
- Developed **Constant Current / Constant Voltage Li-Ion charger (7-cell)** for medical device PCBA redesign
- Hands-on experience in **board bring-up, debugging & failure analysis**
- Conducted **EMI/EMC pre-compliance testing & DFMEA**
- Led team of 3 engineers in validation & verification activities
- Strong exposure to medical (dialysis systems) and automotive electronic systems

PROFESSIONAL EXPERIENCE

Lead Engineer – Electronics

HCLTech, Bangalore | Apr 2024 – Present

Project: PCBA Redesign – PrisMax & Thermax (Kidney Care Medical Devices)

Working on redesign of Power Board and 7-cell Li-Ion CC/CV Charge Controller used in dialysis systems.

Key Contributions:

- Architected board-level hardware design and evaluated alternate design approaches
- Designed constant current / constant voltage charging topology
- Performed component selection, circuit simulation & schematic development
- Conducted PCB design reviews with cross-functional teams
- Executed board bring-up and resolved critical hardware issues during validation
- Participated in EMI/EMC pre-compliance testing
- Led verification activities as per HTPs and compliance standards
- Mentored and led team of 3 engineers in validation and sourcing
- Supported troubleshooting in electromechanical and environmental testing

Hardware Design Engineer

Napino Auto & Electronics, Pune | Dec 2022 – Apr 2024

Project 1: 4kW Battery Management System (48V, 80A) – Automotive

- Defined hardware architecture for high-current BMS
- Designed current sensing, protection, and balancing circuits
- Performed worst-case analysis (WCA) and thermal calculations
- Conducted DFMEA and root cause analysis
- Led board bring-up and validation testing
- Optimized BOM through value engineering, improving cost efficiency
- Coordinated with manufacturing for NPI rollout

Project 2: 3-Speed Seat Ventilation ECU

- Designed and validated control circuitry
- Performed layout review and simulation
- Supported production rollout and troubleshooting

 Awarded *Employee of the Quarter* for leading BMS & ECU design projects.

Executive R&D Engineer

Greaves Cotton Ltd, Aurangabad | Sept 2022 – Dec 2022

- Worked on in-vehicle powertrain integration for BS-6 engine
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R&D Engineer

Star Engineers India Pvt. Ltd., Pune | Dec 2019 – Sept 2022

- Developed schematics using OrCAD
 - Conducted prototype validation and troubleshooting
 - Verified and optimized BOMs
 - Supported customer requirement-based design proposals
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✳ TECHNICAL SKILLS

Core Expertise

- Analog & Power Electronics Design
- Battery Management Systems (BMS)
- DC-DC Converters
- Li-Ion Charging Systems
- Hardware Architecture & Board-Level Design
- EMI/EMC Testing & Compliance
- Design Verification & Validation
- DFMEA, WCA, Root Cause Analysis

Tools & Software

- OrCAD Capture – Advanced
- Altium Designer – Intermediate
- LTspice / PSpice – Advanced
- MATLAB – Intermediate
- Cadence Virtuoso

Communication Protocols

- CAN, LIN, UART, I2C, SPI

Lab Equipment

- DSO, DMM, Function Generator, AC/DC Power Supplies & Electronic Loads
 - ESS/Climatic Chambers, Data Logger
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EDUCATION

B.E. Electrical Engineering

G.H. Rasoni College of Engineering & Technology, Nagpur | 2018 | 84.6%

CERTIFICATIONS

- Electric Vehicle Battery Management System
- MATLAB-SIMULINK Onramp – MathWorks
- Spice Simulation with ORCAD PSpice
- Analog Hardware Design – Hardware Academy
- Learn to Design Your Own Boards – Udemy
- Lt-spice tutorial - Lets simulates Electronics ! - Udemy